

PRESENTATION PITCH / 3 MINUTES

Same journey. Different memories.

A book of light, holding more
than one point of view.

Light Pages

Shape-changing textiles
x shared family journeys



Final concept - AI-assisted visualization

What slips away between destinations?

How might we use **shape-changing textiles** to help families revisit everyday car journeys and discover each other's perspectives, without interrupting their time together?



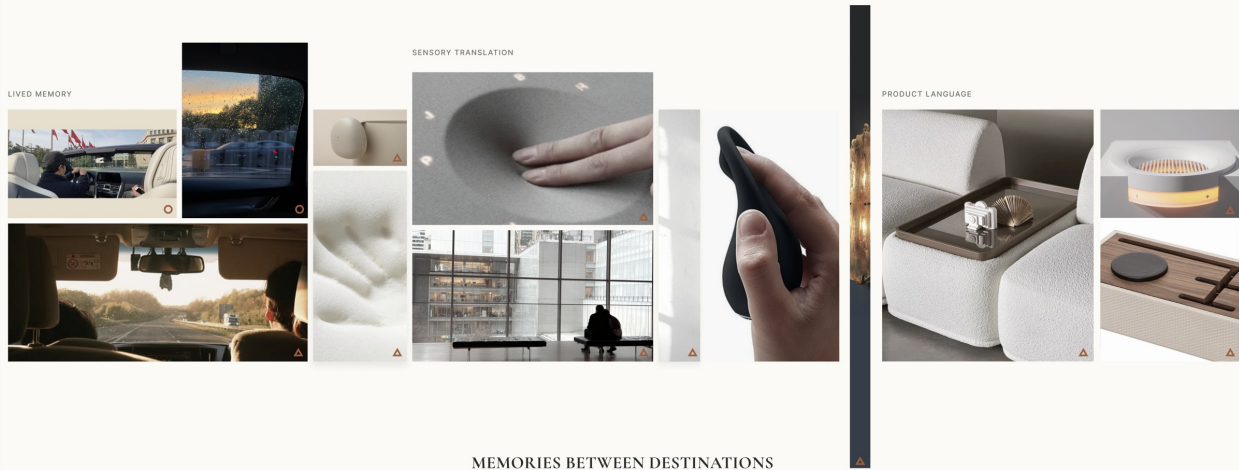
Ordinary shared time, seen from one seat.

MATERIAL AS INTERFACE

Changing textile form proposes a physical entry into a digital memory.
Research direction, not a material used in this prototype.

02 / FROM ATMOSPHERE TO FORM

Why a book of light?



Passing / shared / filtered / portable

A familiar gesture of returning; a new way to notice another perspective.

REARVIEW FOLD
Quick capture → partial choice → memories keeping.

GLANCE
confidence memories

PARK
only opens in P (carpet parking)

REVISIT
one capture memories

SECTION
material roller (camera + glass screen) screen edge slow

Roll Down → CHOOSE
1. rolled inside (during driving)
2. motor rolls screen down
3. flat screen + unlighted bar

REARVIEW FOLD
will no longer press to help small memories from your trip.

Rearview Fold
Capture & revisit

TOGETHER LIGHT
Auto Presence becomes form.

PROTOTYPE TWIST
locked while driving

ARRIVAL
HOME

Together Light
Presence becomes form

WINDOW MEMORY SHADE
Adaptive - reactive, real light, real memories

Filter Glass
Keep the View

Shape-changing Apertures
- tactile press locally change size & shape

Magnetic Edge (Section)
window glass magnetic edge adaptive-reactive

Window Memory Shade
Light becomes a trace

ROAD BOOK LIGHT
Folds flat - opens to gesture

ONE RIDE - ONE NEW CREASE
only one crease for row 11

EACH PAGE - SAME DESIGN
one slim page for each family member

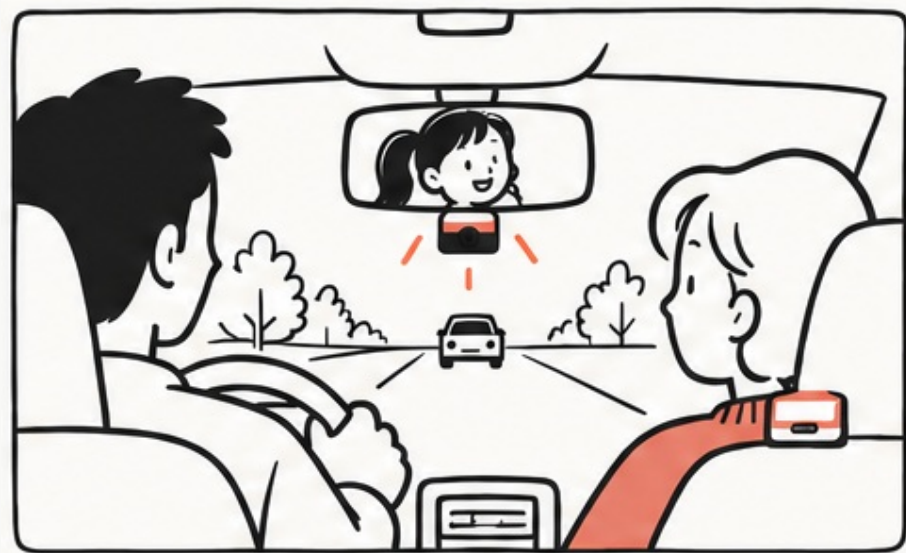
TAKE ONE PAGE
read books quite same journey still with you...

Road Book Light
Selected direction

03 / THE INTENDED EXPERIENCE

Keep. Revisit. Discover.

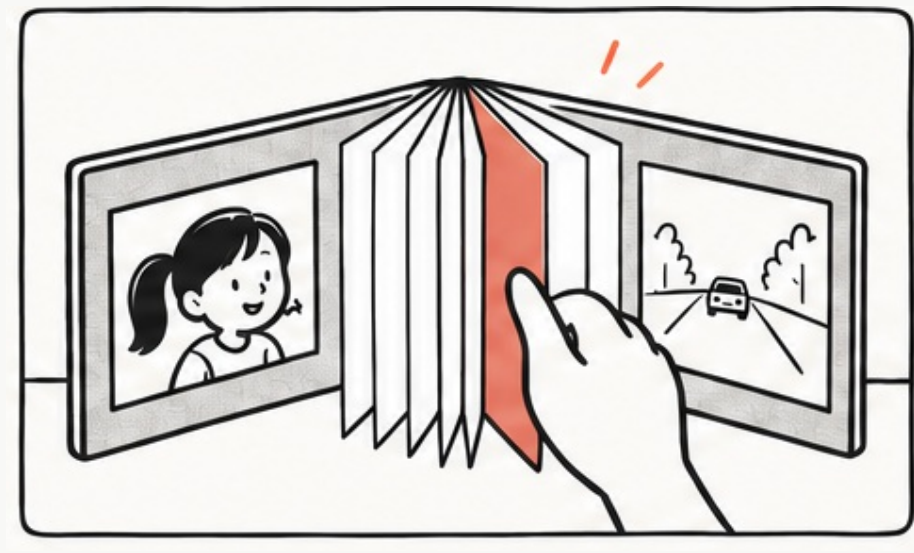
01



Keep a trace

With consent, two modules capture the road and the people travelling together.

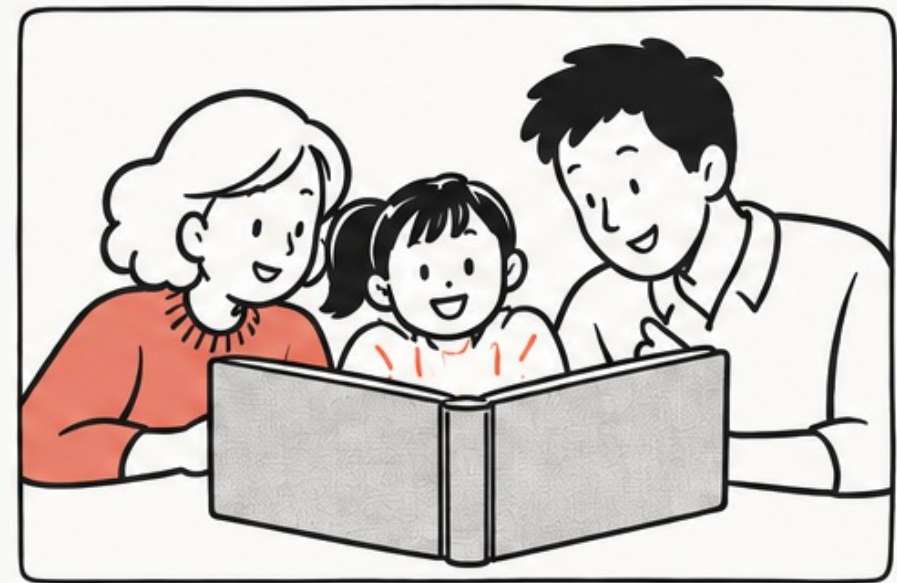
02



Enter through light

At home, opening and turning a page provides a tangible entry into a saved moment.

03



Find another view

Two E-ink images reveal different sides of the same moment, inviting a conversation.

Shape gives a physical entry.

Light cues attention and atmosphere.

Images reveal what we missed.

04 / THE LIVE PROTOTYPE

One light. Another perspective.

Open the physical book by hand, then select a light on the local presentation website.

A / THE PEOPLE WITH US



The moment we might have missed.

B / THE ROAD WE SHARED



The journey we recognise.

THE DEMONSTRATION

1 / Light A

Mirror view

2 / Light B

View from the rear seat

0 / All lights off

Two physical LEDs are individually tested.
Web-to-board integration is prepared and requires
a real-board check after upload.

Classroom prototype: one LED selects one view. Final concept: two E-ink displays show paired views.
Preloaded photos are illustrative, not a synchronized recording. No servo movement in this demonstration.

“I didn’t know you noticed that.”

Can a changing textile page help us
return to a specific shared moment?

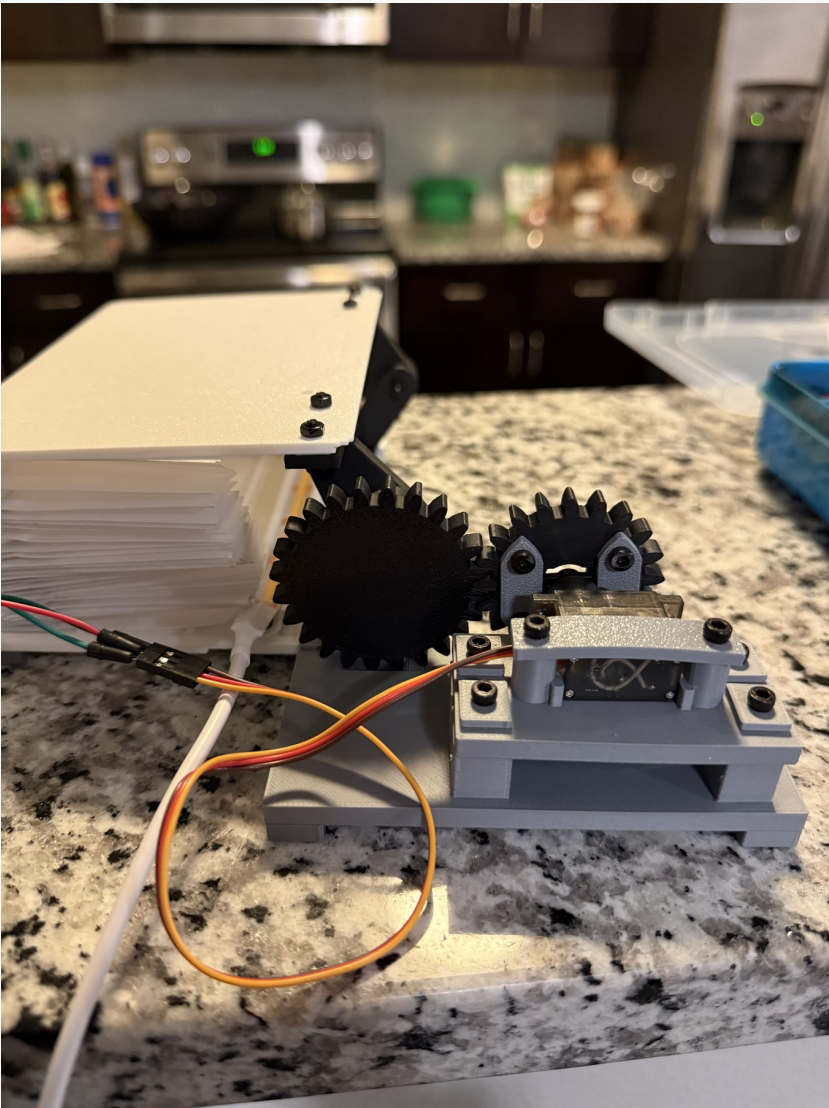
TODAY

A hand-opened book and two
independently tested LEDs. A prepared
photo-to-light interaction.

NEXT

Test textile actuation and reliable capture.
Ask families what the paired views help them
notice.

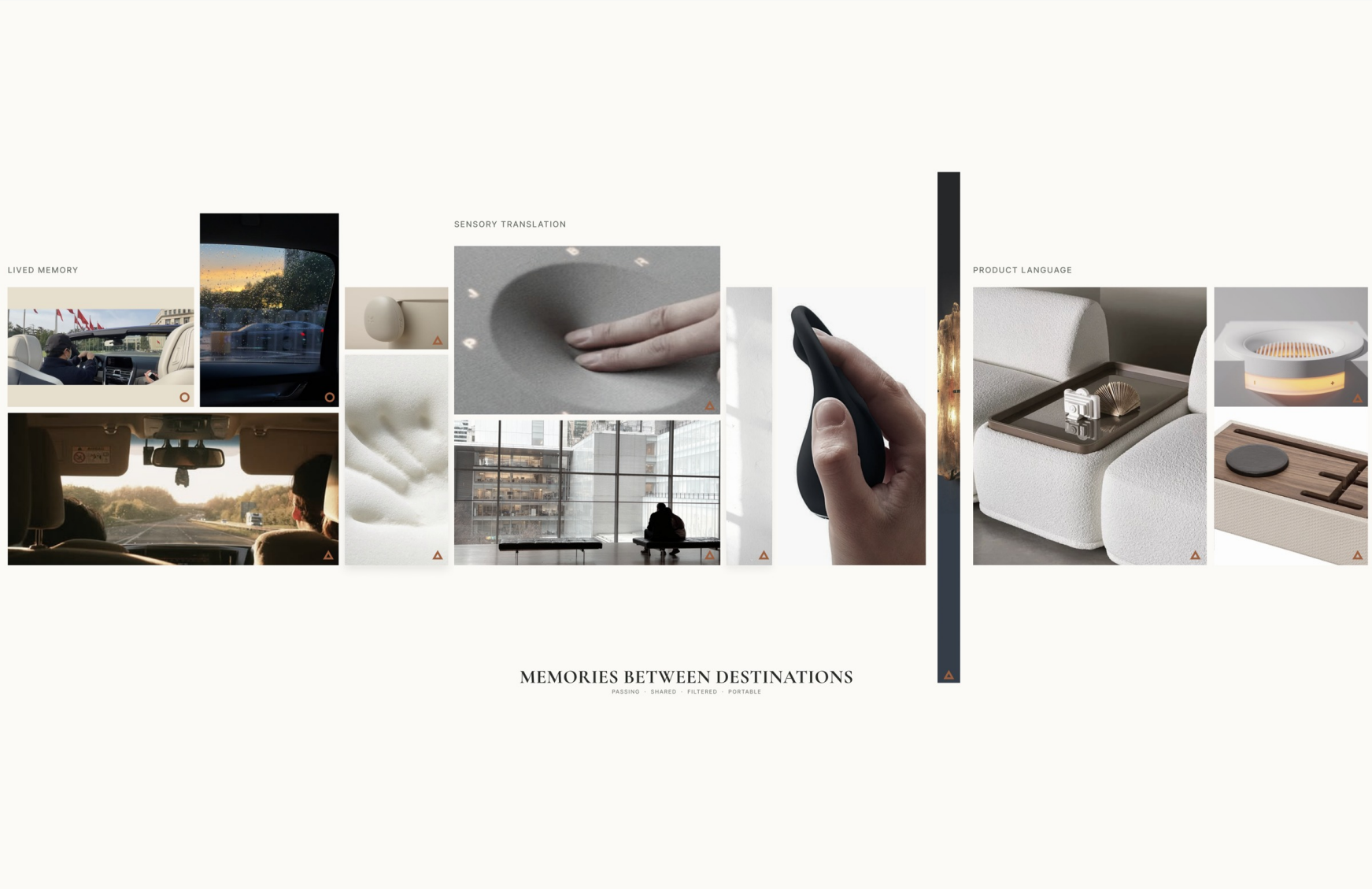
Desired response, not a quote from user testing.



Actual prototype, current build.
Textile actuation and E-ink are not implemented.

APPENDIX 01 / MOODBOARD

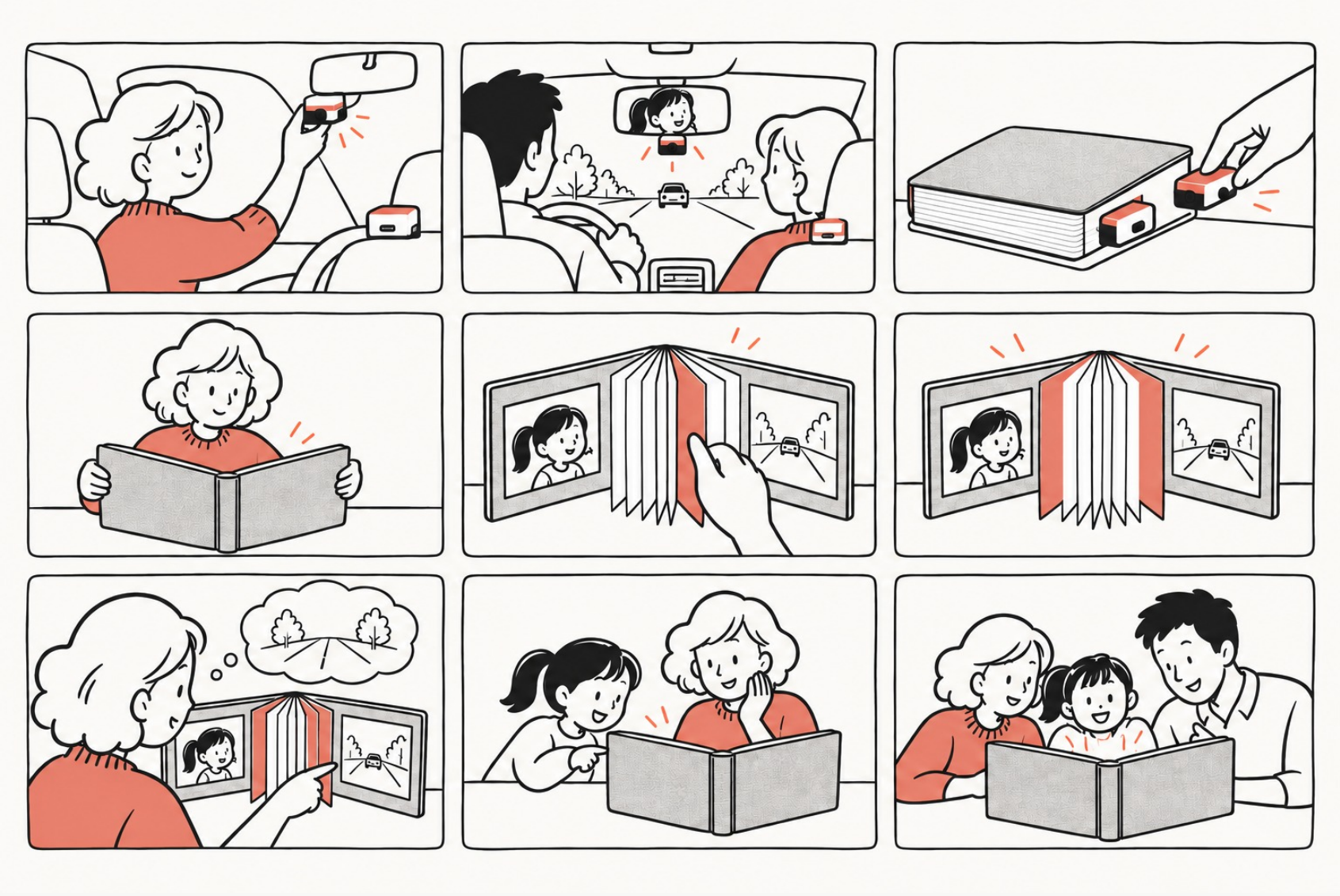
Memories Between Destinations



Final supplied moodboard. Lived memory / sensory translation / product language.

APPENDIX 02 / STORYBOARD

Keep a trace. Enter a memory. Discover another view.



AI-assisted concept storyboard. A proposed family scenario, not evidence of a tested recording system.

Two modules, two directions



Forward-facing module. Rear connector toward the observer.



Cabin-facing module. Lens directed toward rear passengers.

AI-assisted scene edits. Mounting, visibility, crash safety and heat are untested. No driver interaction while moving.

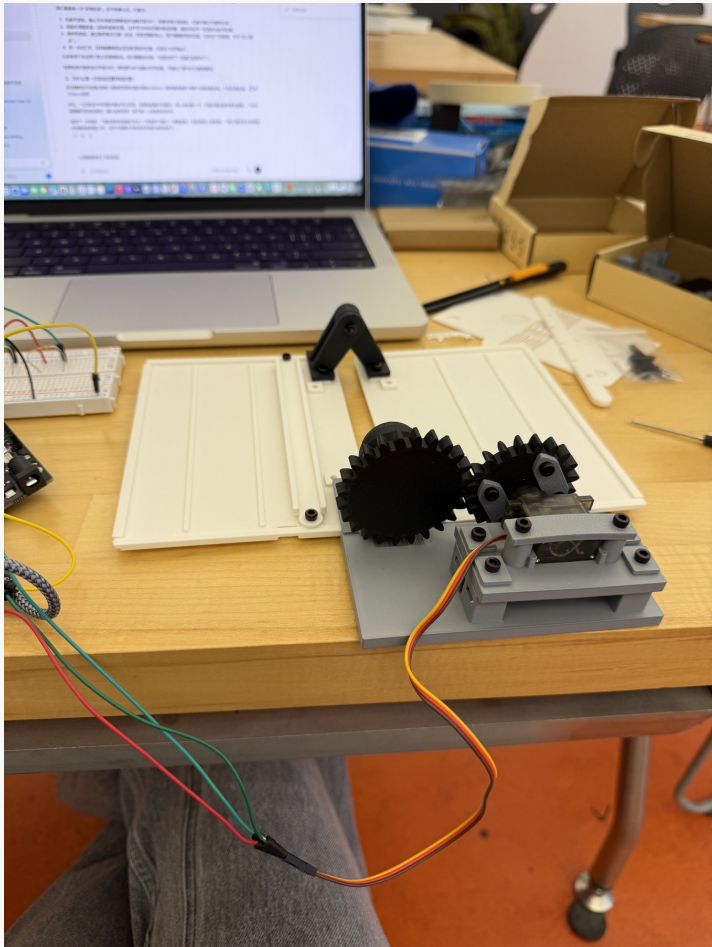
A memory object at home



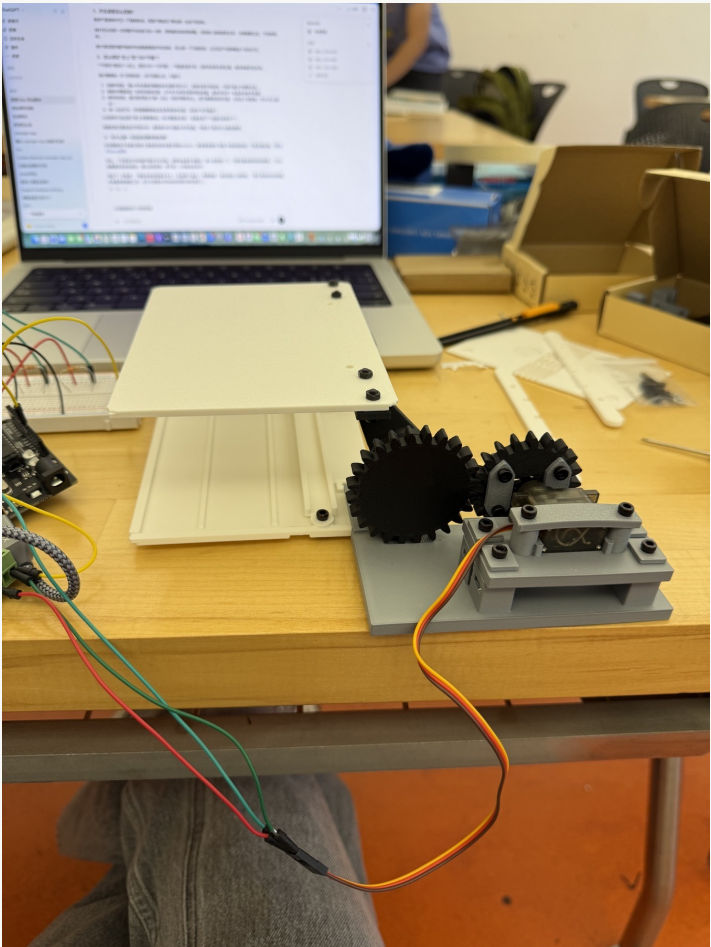
AI-assisted visualization. Fabric covers, light pages, two E-ink concepts and removable modules. Not a manufactured product.

APPENDIX 05 / PHYSICAL ARTIFACT

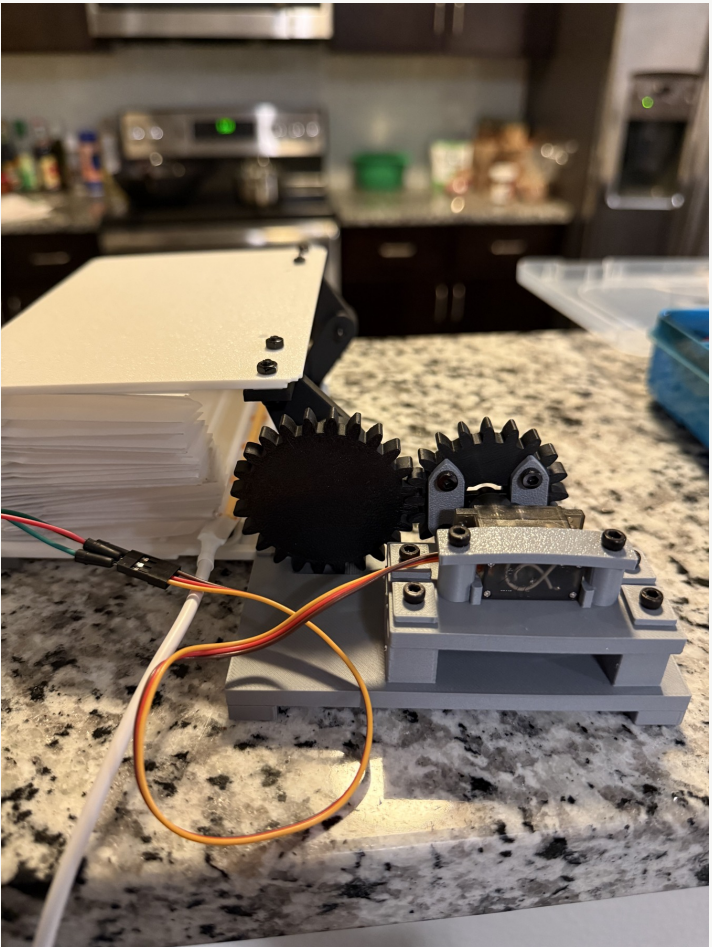
What the prototype actually contains



Earlier build, manually positioned open



Earlier build, closed reference state



Current build with folded pages

The two LEDs have been tested independently. The servo moves unloaded, but powered full opening has not succeeded. Automatic opening is excluded from this presentation. These photographs do not demonstrate a functioning textile actuator.

Shape-changing textiles, honestly situated

RESEARCH REFERENCE

FibeRobo explores temperature-responsive liquid-crystal-elastomer fibres that can be formed into textiles. It provides a reference for form-changing interfaces.

[MIT Media Lab / FibeRobo project overview](#)

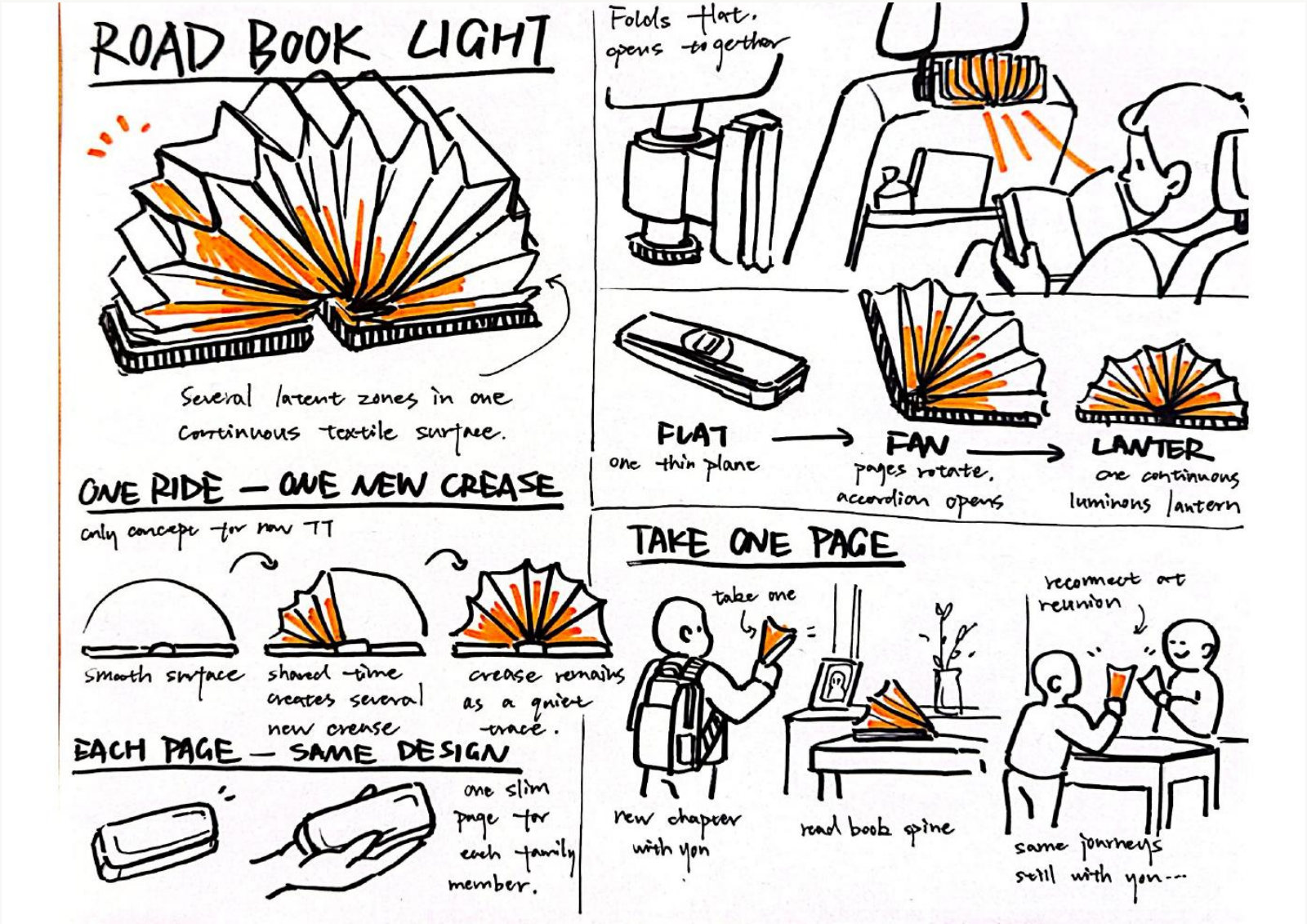
PROPOSED ROLE IN LIGHT PAGES

A textile page changes form to make a selected memory physically noticeable. Light cues attention; paired images provide context. Actuator placement, heat, power, response time and repeatability remain open design questions.

CURRENT PROTOTYPE BOUNDARY

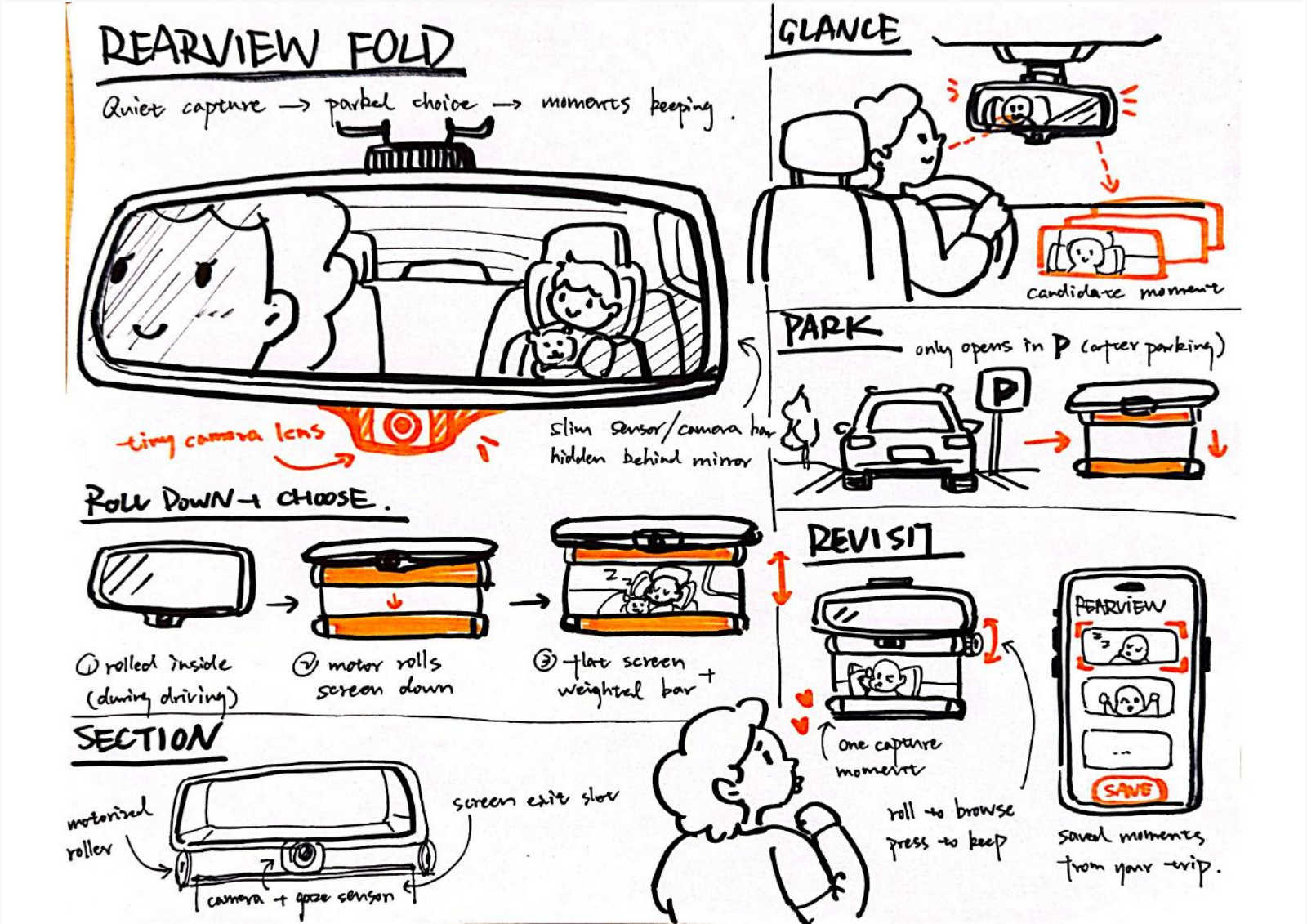
- Manual opening + two LEDs + preloaded photographs.
- No working shape-changing fibre, actual E-ink or trip archive.
- No verified synchronization, vehicle-ready mounting or proven benefit to memory.
- The material proposal is an emerging-technology direction, not a claim about the current build.

Road Book Light



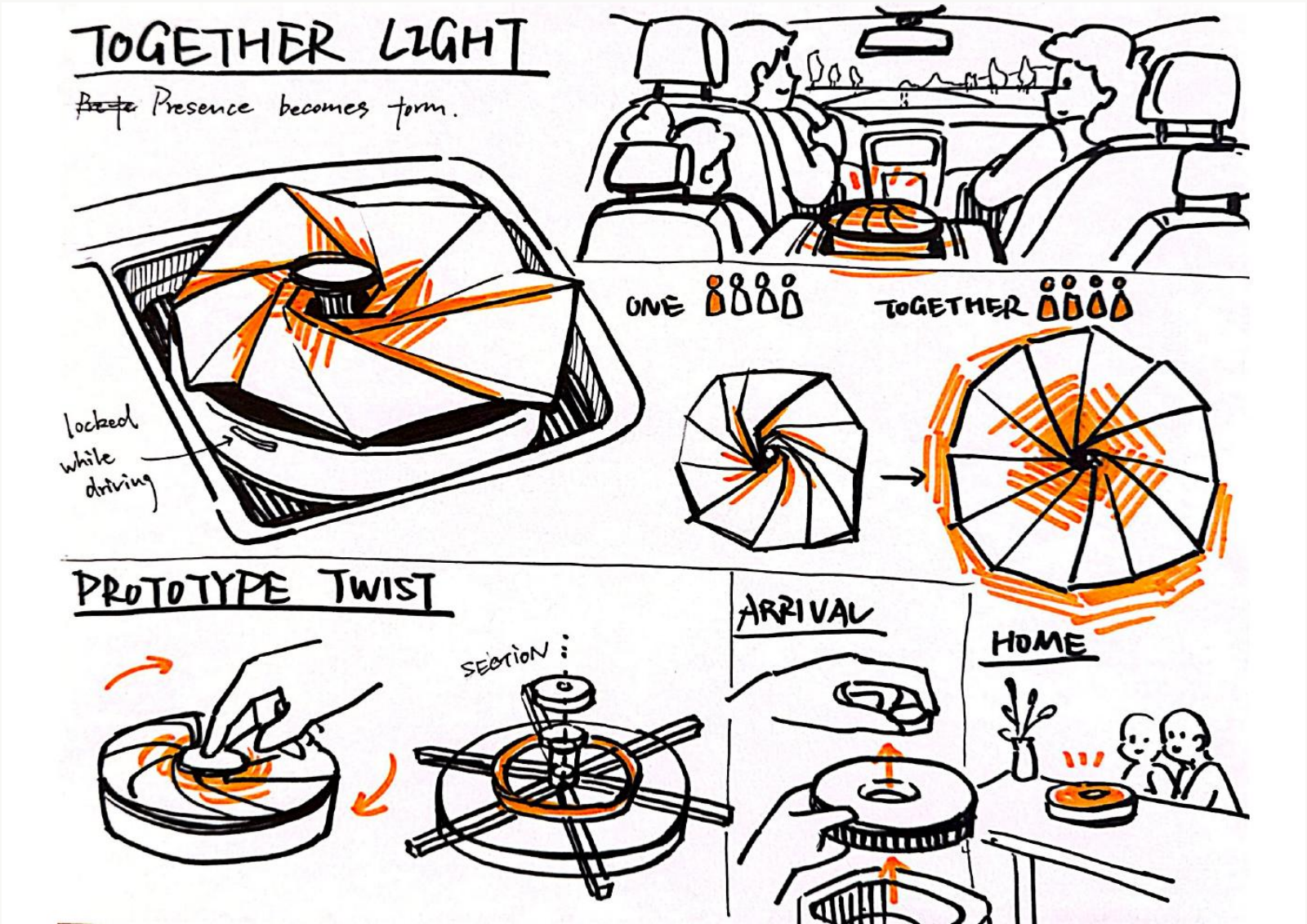
Original supplied scan. Exploratory proposal; mechanisms, safety and material behaviour have not been validated.

Rearview Fold



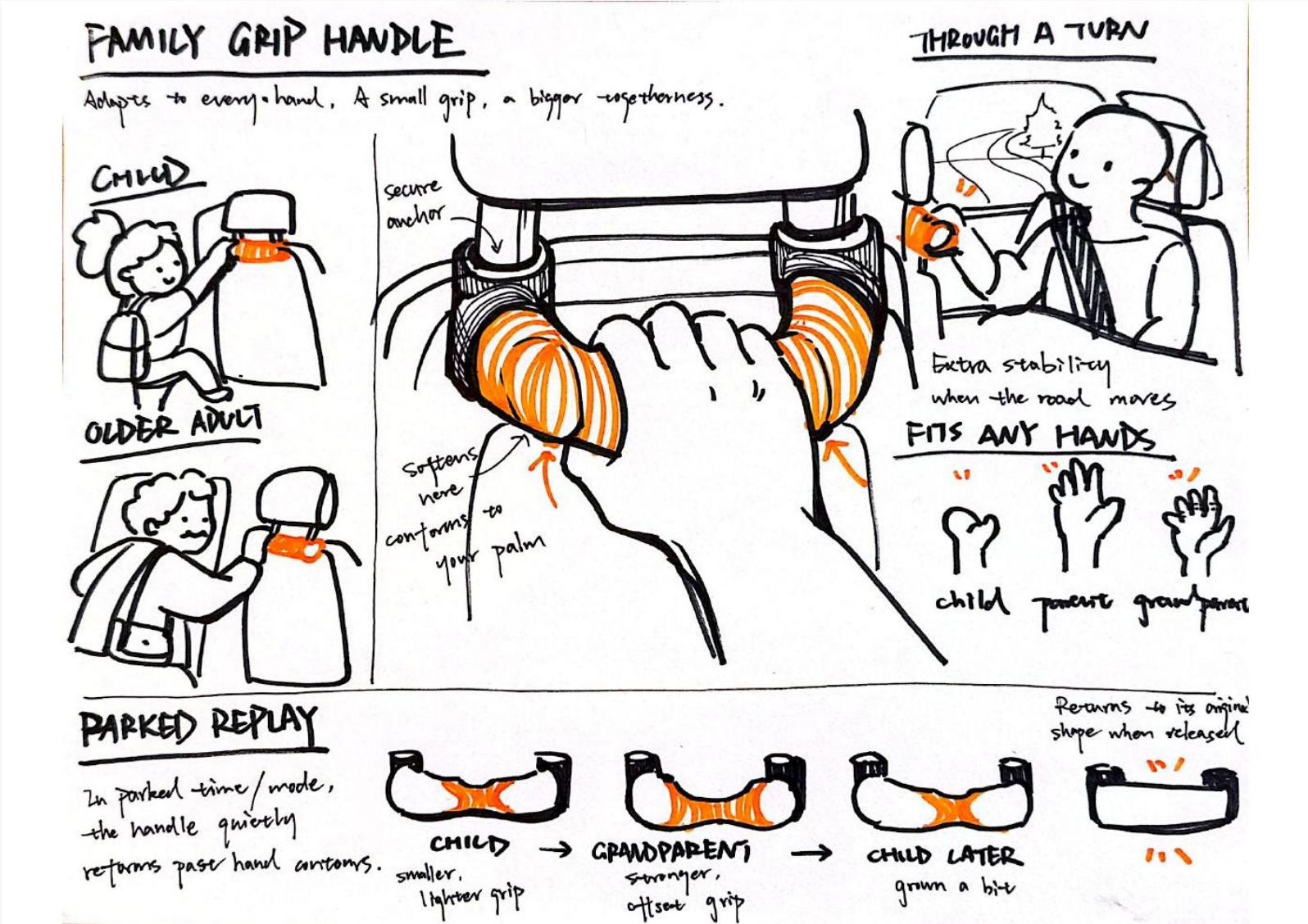
Original supplied scan. Exploratory proposal; mechanisms, safety and material behaviour have not been validated.

Together Light



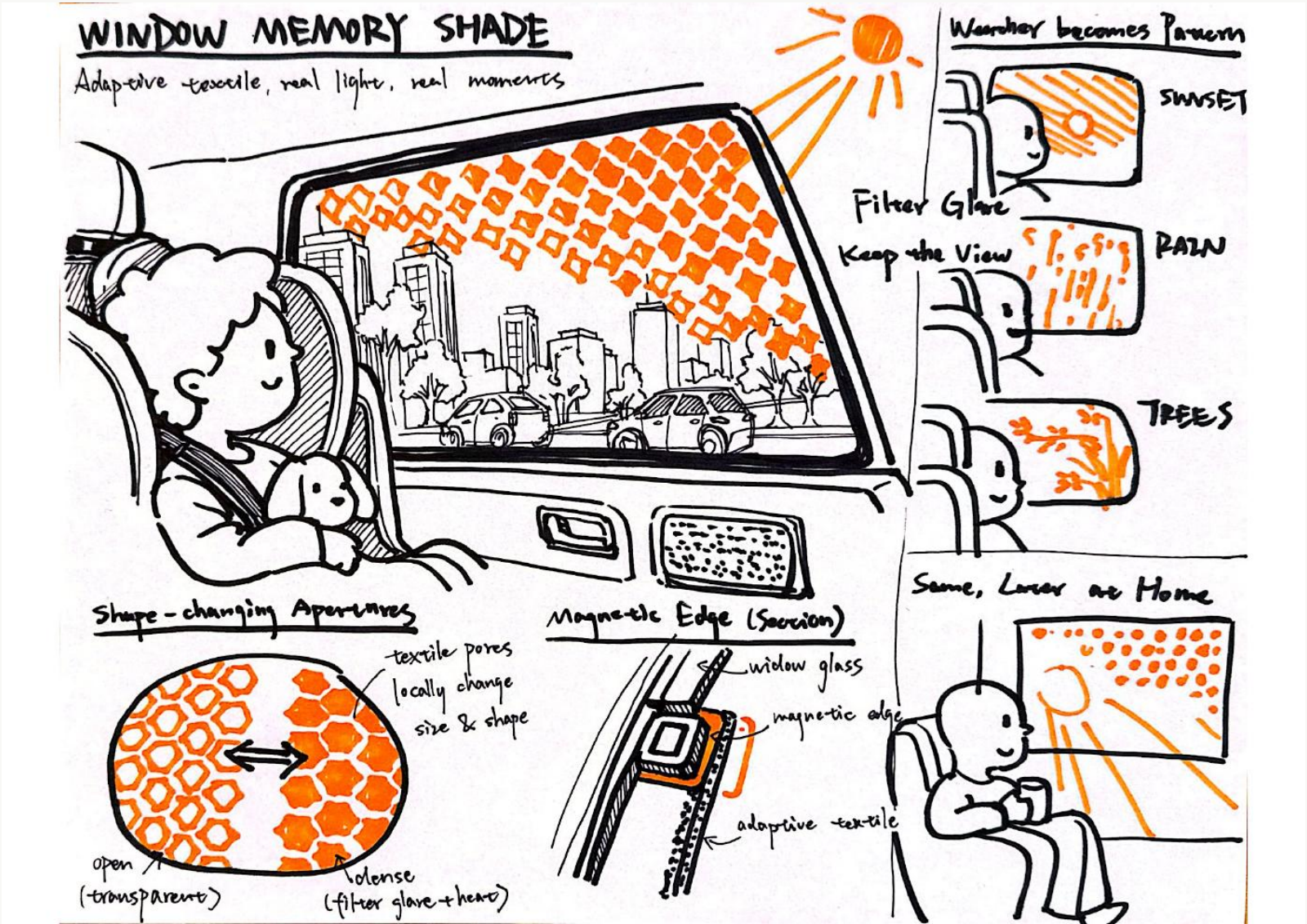
Original supplied scan. Exploratory proposal; mechanisms, safety and material behaviour have not been validated.

Family Grip Handle



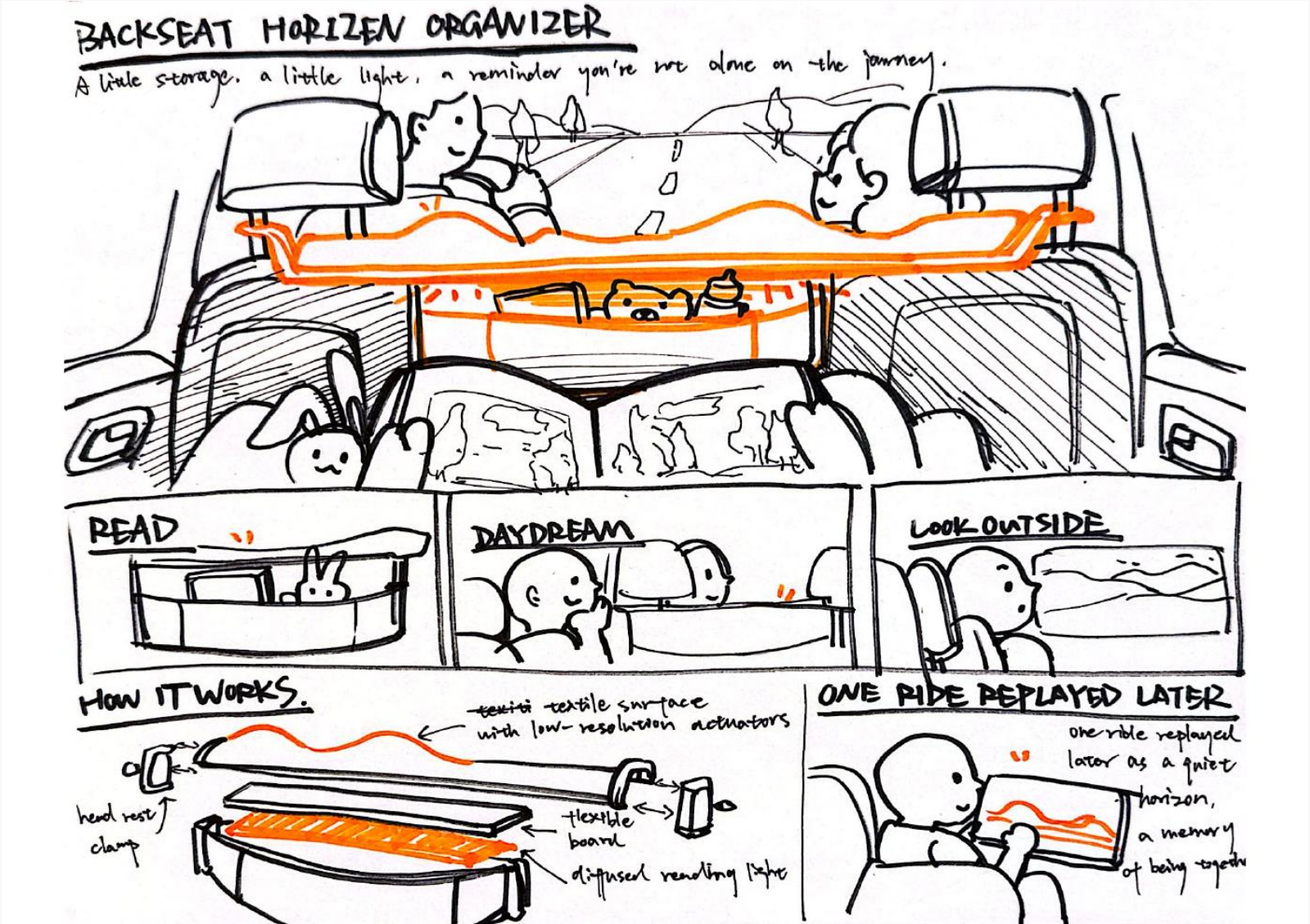
Original supplied scan. Exploratory proposal; mechanisms, safety and material behaviour have not been validated.

Window Memory Shade



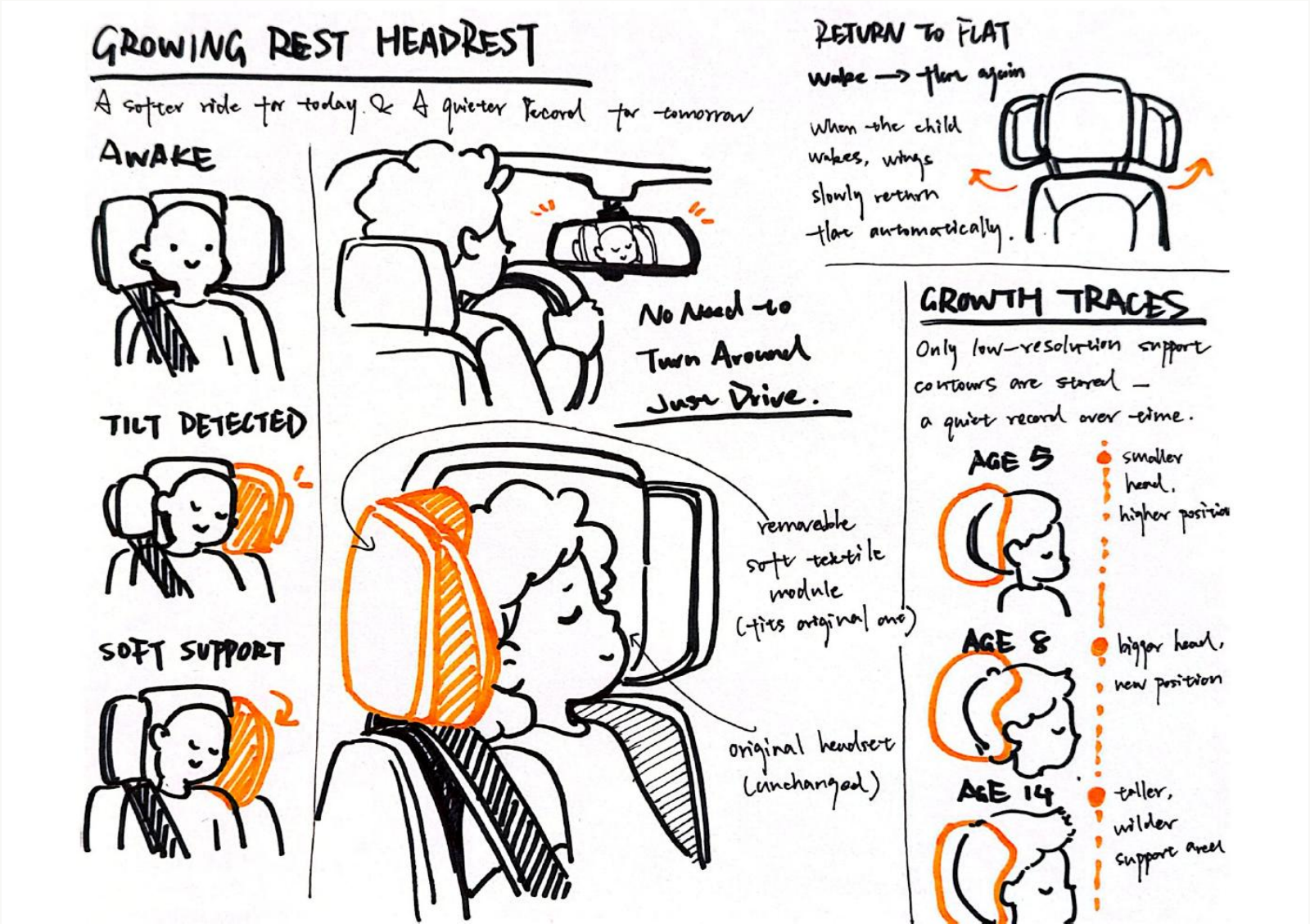
Original supplied scan. Exploratory proposal; mechanisms, safety and material behaviour have not been validated.

Backseat Horizon Organizer



Original supplied scan. Exploratory proposal; mechanisms, safety and material behaviour have not been validated.

Growing Rest Headrest



Original supplied scan. Exploratory proposal; mechanisms, safety and material behaviour have not been validated.

Evidence, credits and the remaining photograph

PROJECT MATERIAL

Final moodboard and all seven scanned sketch pages supplied by Yi (Levy) Liu. Concept renders, edited car scenes and storyboard are AI-assisted visualizations. Physical prototype photographs document the build.

The two reference photographs illustrate different viewpoints. They are not asserted to have been captured during one synchronized journey. Moodboard references remain references, not original product work.

[MIT Media Lab: FibeRobo research reference](#)

[Chrome: Web Serial implementation reference](#)

REQUIRED BEFORE FINAL SUBMISSION

Add the photograph of your desk with feedback Post-its after the presentation.

This requested photograph and actual class feedback have not yet been supplied. They are deliberately not fabricated or represented by a substitute.

Submission specified by the instructor:
Presentation Pitch + appendix (PDF)
Sep 30, 5 pm.